

UCE857: INTEGRATED WATERSHED MANAGEMENT				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To impart students with a comprehensive understanding of watershed systems and need for a holistic approach to managing a watershed, considering all aspects of the ecosystem				
Introduction: Definition, principles, classification and geomorphology of watersheds, physical and hydrological characteristics of watershed, factors affecting watershed management				
Fundamentals of watershed hydrology: Understanding the natural processes within a watershed, water flow dynamics, rainfall-runoff modelling and water yield assessments from watershed				
Soil, land use and water quality assessment and management: Principles of soil erosion, estimation of soil erosion using soil erosion models, land cover mapping, point and non-point pollution parameters, streamflow measurements. Strategies to mitigate soil erosion including contour farming, terracing, check dams, and vegetation restoration, best management practices.				
Integrated planning and decision making: Approaches to develop comprehensive watershed management plans considering social, economic, and environmental factors, stakeholder engagement and collaborative decision-making, introduction to multi criteria decision making and modelling tools in a watershed.				
Course Learning Objectives (CLO) The students will be able to: 1. Define a watershed as a system and identify major environmental problems within a watershed. 2. Perform soil, land use and water quality assessments within a watershed 3. Incorporate stakeholder engagement and community participation in decision-making processes.				
Text Books: 1. Debarry Paul A.. <i>Watershed: Processes, Assessment and Management</i>, John Wiley & Sons, New Jersey, 2004 2. Dhruva N.V.V., Sastry G. and Patnaik U.S. 1990. <i>Watershed Management</i>, Indian Council of Agricultural Research, New Delhi, 1990 3. Murthy, J.V.S., <i>Watershed Management</i>, New Age International Publishers, 2017 4. Iyer K. G. and Roy U.N., (ed.). <i>Watershed Management and Sustainable Development</i>, Kanishka Publishers, New Delhi, 2005 Reference Books: 1. Isobel W. Heathcote., (2009) <i>Integrated Watershed Management: Principles and Practice</i>. Wiley & Sons, Incorporated, John, USA, second edition, 2009 2. Tideman E.M. <i>Watershed Management–Guidelines for Indian Conditions</i>, Omega Scientific Publishers, New Delhi, 1999 3. Paul A. DeBarry. (2004). <i>Watersheds: Processes, Assessment and Management</i>. Wiley & Sons, Incorporated, John, USA, second edition, 2004 4. Randhir O. Timothy, 2007. <i>Watershed Management-Issues and Approaches</i>, IWA Publishing, 2007				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25